Vinayaka Sajjanshetty

+81-7084689061 hellovinayakasajjanshetty@gmail.com Hamamatsu, Japan
linkedin.com/in/vinayaka-sajjanshetty github.com/vinayakas1997 vinayakas1997.github.io

Summary

Forward Deployed AI & Systems Engineer with 4+ years of full-stack experience delivering production AI agentic workflows and edge infrastructure. Specialized in multi-agent orchestration (LangGraph), LLM evals (Golden Datasets), on-premise system scoping, and real-time factory telemetry.

Technical Skills

AI & Agentic Systems: Multi-Agent Orchestration (LangGraph), Enterprise RAG, LLM Evals (Golden Datasets), Guardrails (AST/Regex), Knowledge Graphs (Hindsight), vLLM, Atlas Inference Engine
Languages & Core: Python, C/C++, TypeScript, React, FastAPI, PyTorch, SQL, Docker, Linux, Git, ROS2
Systems & IIoT: OPC UA, PostgreSQL, Prometheus, Grafana, Ansible, FINS Protocol, LINX TRITON Edge, Jetson Nano
Spoken Languages: English (Fluent), Japanese (Business / Active Technical Collaboration), Hindi (Native)

Experience

Somic Ishikawa — Shizuoka, Japan
Applied AI & Systems Engineer (R&D) — *Oct 2024 – Present*

- Led end-to-end technical discovery and production rollout of enterprise AI systems on factory floors; served as primary US–Japan liaison, resolving data bottlenecks and aligning field feedback into model roadmaps.
- Authored executive feasibility study evaluating cloud SaaS migration to an air-gapped Gitea + PostgreSQL stack; benchmarked 6 platforms across compute constraints and scoped GPU trade-offs for a local LLM reviewer.

Legato Health Technologies — Bangalore, India
Software Engineer, Data & Analytics — *Sep 2020 – Sep 2022*

- Partnered with cross-functional healthcare stakeholders to architect scalable, backward-compatible ETL data models handling patient analytics, reducing production data anomalies by 25% via automated validation.

Bosch Automotive Electronics — Bangalore, India
Engineering Intern — *Jan 2019 – Mar 2019*

- Investigated thermal manufacturing defects and developed 3 real-time CV monitoring models (OpenCV), identifying optimal operating windows and accelerating defect diagnostics by 30%.

Projects

Industrial IIoT & Multi-Agent Factory Analytics Platform [Phase 1] [Phase 2] [GitHub] — 2024 – Present

- **Phase 1 (Edge Ingestion & Cost Optimization):** Eliminated high-cost legacy IoT licenses and data rate caps by building an open edge-to-PostgreSQL pipeline on LINX TRITON controllers (OPC UA/MQTT/FINS); unlocked unrestricted shop-floor data access for analytics teams and set up Grafana/Prometheus telemetry.
- **Phase 2 (Dual-User AI Agentic Analytics):** Deployed a LangGraph multi-agent system delivering dual-purpose intelligence: (1) real-time shift feasibility commentary for plant operators near machines (predicting plan achievability & downtime), and (2) autonomous deep-research analysis for engineering teams with self-correcting queries and chart generation.

Intelligent RAG with Knowledge Graph [Phase 1] [Phase 2] — 2024 – Present

- **Platform Adoption & Cross-Team Scaling:** Engineered an on-premise autonomous RAG platform over 2,500+ bilingual (JP/EN) docs, accelerating query synthesis 360x (hours to <10s); scaled platform adoption across internal PLC engineering and business teams with domain-specific guidelines.
- **VLM Reliability & Hardware Optimization:** Solved 45% OCR failure rates via MinerU layout parsing with 3-tier VLM fallback; doubled on-prem inference throughput (35 → 80 tok/s via Atlas on DGX Spark, 90s → 6s latency) and built Phase 2 HITL Eval Dashboard achieving 85–90% retrieval precision.

Parallel Multi-Agent 1:1 Corporate Interviewer [Case Study] [GitHub] — 2024 – Present

- **Aim & Business Impact:** Designed an autonomous interview platform to eliminate corporate meeting fatigue; replaced 10+ hours of manual manager 1:1s with parallel AI conversational streams, achieving a **10x time reduction (<1 hr)** and unbiased, anonymized team sentiment collection.
- **System Architecture:** Built a 4-agent state machine (Planner → Interviewer → Extractor → Synthesis) with adaptive dialogue logic that **intelligently analyzes user responses in real time, smartly reframes follow-up questions**, and synthesizes executive reports.

Education

Shizuoka University — Shizuoka, Japan
Master of Science in Computer Science — *Oct 2022 – Sep 2024*

PES University — Bangalore, India
Bachelor of Engineering in Electronics & Communication — *Jul 2015 – Jun 2019*